CHAPTER 5

Elasticity & Its Application

INTRO — What Is Elasticity?

EASY DEFINITION

Elasticity is a number that tells us HOW MUCH buyers or sellers react when something changes — like a price change or an income change.

Think of a rubber band. Stretch it a lot → elastic. Barely moves → inelastic.

In economics: if price changes, does quantity demanded change a LOT or a LITTLE? That answer is elasticity.

REAL-LIFE ANALOGY

PETROL: Price goes up 20%. You still need to drive. You buy almost the same amount → INELASTIC (not very responsive).

RESTAURANT MEALS: Price goes up 20%. You cook at home instead. Demand falls a lot → ELASTIC (very responsive).

WHY THIS CHAPTER MATTERS

Chapter 4 told us the direction of change (demand down when price up). Chapter 5 tells us the SIZE of that change. Elasticity is the measuring tool.

By the end you will answer: Why can good farming news be bad for farmers? Why did OPEC fail? Does drug interdiction reduce crime?

5-1 Price Elasticity of Demand (PED)

EASY DEFINITION

Price Elasticity of Demand (PED) measures how much the QUANTITY DEMANDED of a good responds when its PRICE changes.

Price up 10%, quantity drops 20% → big response → ELASTIC demand.

Price up 10%, quantity drops 2% → small response → INELASTIC demand.

SIMPLE EXAMPLE

You sell ice cream. You raise price from $2 to $4 (price doubles). If most customers walk away → ELASTIC. If most still buy → INELASTIC.

5-1a The 4 Determinants of PED

KEY IDEA

These four factors explain WHY demand for some goods is elastic while demand for others is inelastic. All four must be memorized for the exam.

The 4 Determinants — Summary Table

#	Factor	MORE Elastic When...	MORE Inelastic When...
1	Availability of Substitutes	Many close substitutes exist (butter → use margarine)	Few/no substitutes (eggs, insulin)
2	Necessity vs. Luxury	Luxury goods (sailboats, diamonds) — can skip them	Necessities (medicine, bread) — must have them
3	Market Definition	Narrowly defined (vanilla ice cream — many flavors replace it)	Broadly defined (food — nothing replaces all food)
4	Time Horizon	Long run — time to change habits, buy different cars	Short run — habits don't change overnight

MEMORY TRICK — S·N·M·T

S = Substitutes | N = Necessity/Luxury | M = Market Definition | T = Time Horizon
Say: "Sellers Need Market Time" — four words, four factors!

REAL-LIFE EXAMPLES

Substitutes: Pepsi and Coke are close substitutes. If Coke price rises, people switch to Pepsi → elastic demand for Coke.
Necessity: Rice in Bangladesh — eaten every meal, price barely stops people → inelastic. Diamonds → luxury → elastic.
Market definition: 'Food' (broad) = inelastic. 'McDonald's Big Mac' (narrow) = elastic (many fast-food alternatives).
Time: Gasoline — 10% price rise reduces consumption ~2.5% after 1 year, but ~6% after 5 years (people buy fuel-efficient cars).

EXAM KEY POINTS

Whether a good is a necessity or luxury depends on BUYER PREFERENCES, not the product itself.
An avid sailor may view sailboats as a necessity and doctor visits as a luxury.
Vanilla ice cream is elastic — other flavors are near-perfect substitutes.
The broad category 'food' is inelastic — nothing replaces all food overall.

5-1b Computing Price Elasticity of Demand

📐 **MAIN FORMULA**

PED = % Change in Quantity Demanded ÷ % Change in Price

PED Price Elasticity of Demand — the number we calculate
%ΔQd Percentage change in quantity demanded
%ΔP Percentage change in price

▨ **BOOK EXAMPLE**

Ice cream price rises 10%. Quantity demanded falls 20%.
PED = 20% ÷ 10% = 2
Quantity responded TWICE as much as price → Elastic demand (PED > 1)

⚠ **ABOUT THE MINUS SIGN**

Price and quantity always move in opposite directions (law of demand), so PED is technically negative.
Economists DROP the minus sign and report PED as a positive number.
A HIGHER number = MORE elastic. Don't panic about the negative sign.

Step-by-Step: How to Calculate PED

1	**Find % change in Quantity Demanded** (New Qty − Old Qty) ÷ Old Qty × 100
2	**Find % change in Price** (New Price − Old Price) ÷ Old Price × 100
3	**Divide** PED = %ΔQd ÷ %ΔP (drop the minus sign from result)
4	**Interpret** PED > 1 → Elastic | PED < 1 → Inelastic | PED = 1 → Unit Elastic

💡 **WORKED EXAMPLE**

Coffee price rises from $2 to $3. Quantity demanded falls from 100 to 70 cups.
Step 1 — %ΔQd = (70−100) ÷ 100 × 100 = −30% → drop minus → 30%
Step 2 — %ΔP = (3−2) ÷ 2 × 100 = 50%
Step 3 — PED = 30% ÷ 50% = 0.6
Step 4 — PED = 0.6 → Less than 1 → INELASTIC demand

5-1c The Midpoint Method

THE PROBLEM & SOLUTION

THE PROBLEM: Going from Point A→B gives a different PED than B→A. This is inconsistent.
A→B: Price +50%, Qty −33% → PED = 0.66
B→A: Price −33%, Qty +50% → PED = 1.5
Same two points — two different answers! This is the problem.

THE SOLUTION — Midpoint Method: Use the AVERAGE (midpoint) of the two values as the base for % calculations. Always gives the same answer in both directions.

MIDPOINT METHOD FORMULA

$$PED = [(Q_2-Q_1) / ((Q_2+Q_1)/2)] \div [(P_2-P_1) / ((P_2+P_1)/2)]$$

Q_1, P_1 Original quantity and price (starting point)
Q_2, P_2 New quantity and price (ending point)
$(Q_1+Q_2)/2$ Midpoint = average of the two quantities (use as base)
$(P_1+P_2)/2$ Midpoint = average of the two prices (use as base)

Step-by-Step: Midpoint Calculation (Book Example — Points A & B)

1 **Identify both points**
A: P=$4, Q=120 | B: P=$6, Q=80

2 **Midpoint price**
(4+6) ÷ 2 = $5

3 **Midpoint quantity**
(120+80) ÷ 2 = 100

4 **%ΔP**
(6−4) ÷ 5 × 100 = 40%

5 **%ΔQd**
(80−120) ÷ 100 × 100 = −40% → drop minus → 40%

6 **PED = 40% ÷ 40% = 1.0 (Unit Elastic)**
Going B→A also gives 40% ÷ 40% = 1.0 ✓ Same answer both directions!

BOOK MCQ PRACTICE — MIDPOINT METHOD

Q: Price rises from $8 to $12. Qty demanded falls from 110 to 90. What is PED?
Midpoint P = (8+12)/2 = 10 | %ΔP = (12−8)/10 × 100 = 40%
Midpoint Q = (110+90)/2 = 100 | %ΔQ = (90−110)/100 × 100 = 20%
PED = 20% ÷ 40% = 0.5 = 1/2 → INELASTIC (Book answer: choice b)

MOST COMMON MISTAKE

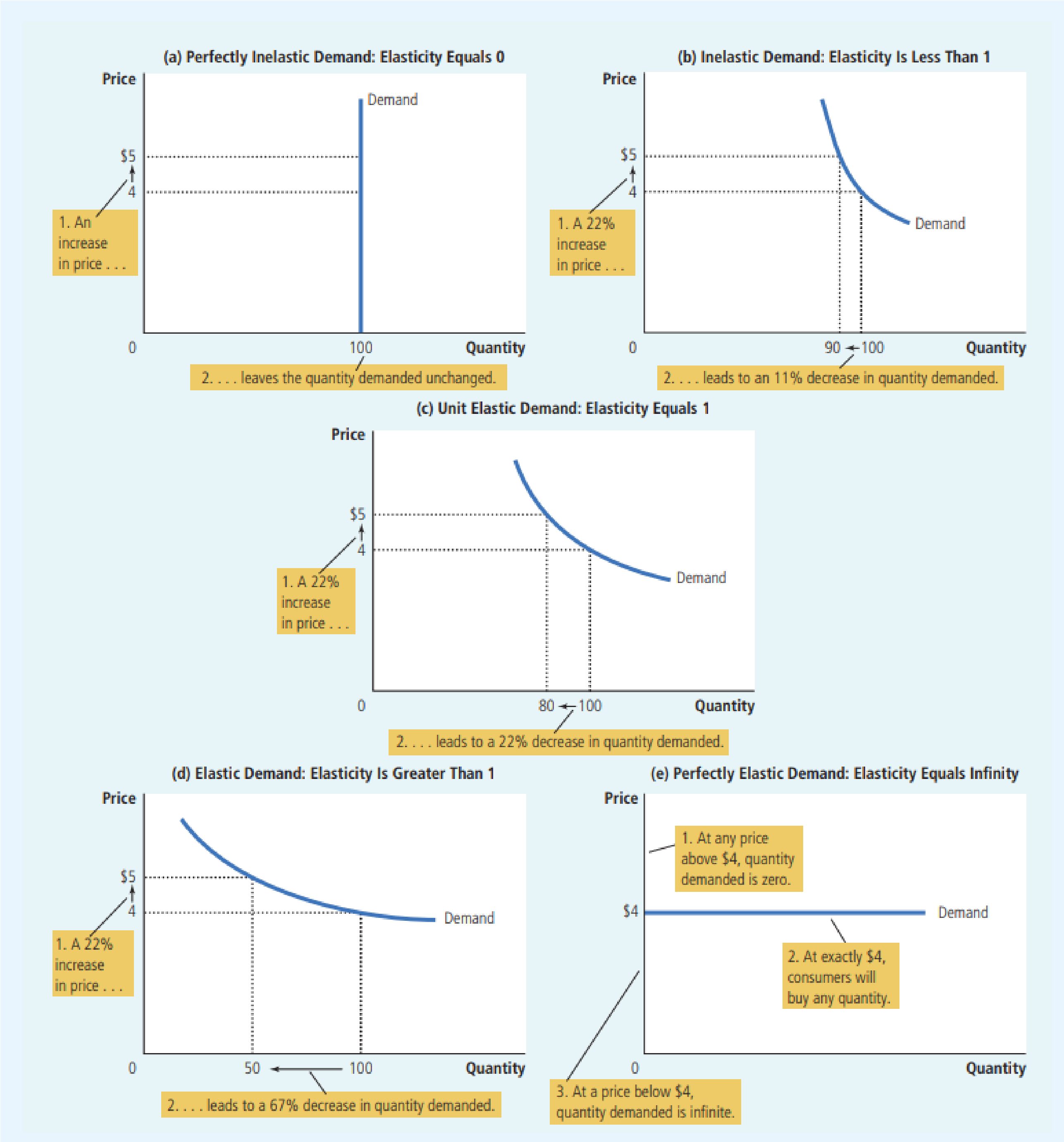
Students use the OLD value as the base instead of the midpoint (average) → gets wrong answer.
Always divide by the AVERAGE of the two values when using the midpoint method.

5-1d The 5 Types of Demand Curves

KEY RULE

FLATTER demand curve → MORE elastic. | STEEPER demand curve → MORE inelastic.
There are 5 types, ranging from perfectly inelastic (vertical line) to perfectly elastic (horizontal line).

Type	Elasticity	Curve Shape	What It Means	Example
Perfectly Inelastic	E = 0	Vertical line ↕	Any price change → quantity unchanged	Life-saving insulin
Inelastic	E < 1	Steep downward	Price up a lot → qty falls a little	Cigarettes, petrol, rice
Unit Elastic	E = 1	Moderate slope	Price & qty change by same %	Some manufactured goods
Elastic	E > 1	Gentle/flat slope	Small price rise → big qty drop	Restaurant meals, luxury cars
Perfectly Elastic	E = ∞	Horizontal line ↔	Any price rise → qty falls to zero	Perfectly competitive market



Real-World Price Elasticities (Book FYI Box)

Good	PED Value	Type
Eggs	0.1	Very Inelastic
Healthcare	0.2	Very Inelastic
Cigarettes	0.4	Inelastic
Rice	0.5	Inelastic

Housing	0.7	Inelastic
Beef	1.6	Elastic
Peanut Butter	1.7	Elastic
Restaurant Meals	2.3	Very Elastic
Mountain Dew (soda)	4.4	Highly Elastic

MEMORY TRICK (FROM THE BOOK ITSELF!)

"Inelastic curves look like the letter I" — this is literally stated in Mankiw's textbook.
A vertical demand curve = perfectly inelastic = shaped like the letter I.
Use this on your next exam whenever asked about curve shapes!

5-1e Total Revenue & Price Elasticity of Demand

WHAT IS TOTAL REVENUE?

Total Revenue (TR) = Price × Quantity (P × Q)
It is the total money that buyers pay and sellers receive.
On a demand graph, TR is the RECTANGLE area under the price line (height = P, width = Q).
Book example: P = \$4, Q = 100 → TR = \$4 × 100 = \$400

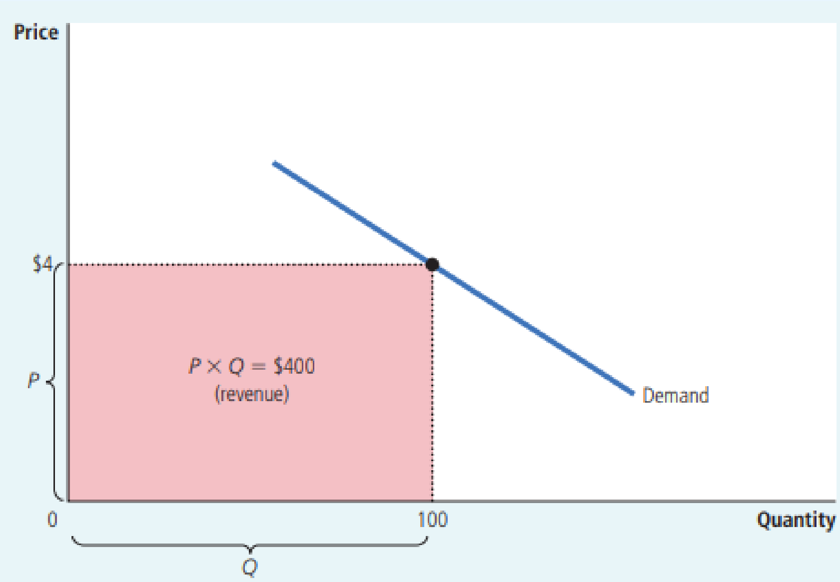


FIGURE 2

Total Revenue

The total amount paid by buyers, and received as revenue by sellers, equals the area of the box under the demand curve, $P \times Q$. Here, at a price of \$4, the quantity demanded is 100 and total revenue is \$400.

The Golden Rules — Total Revenue & Elasticity

Demand Type	PED	Price RISES → TR...	Price FALLS → TR...
Inelastic	< 1	RISES ↑ (same direction)	FALLS ↓
Unit Elastic	= 1	UNCHANGED →	UNCHANGED →
Elastic	> 1	FALLS ↓ (opposite direction)	RISES ↑

WHY DOES THIS HAPPEN? THE LOGIC

INELASTIC (e.g., medicine): You raise price. People still buy nearly the same amount. The 'price gain' is bigger than the 'quantity loss.' → TR goes UP.

ELASTIC (e.g., restaurant): You raise price. Many customers leave. The 'quantity loss' is bigger than the 'price gain.' → TR goes DOWN.

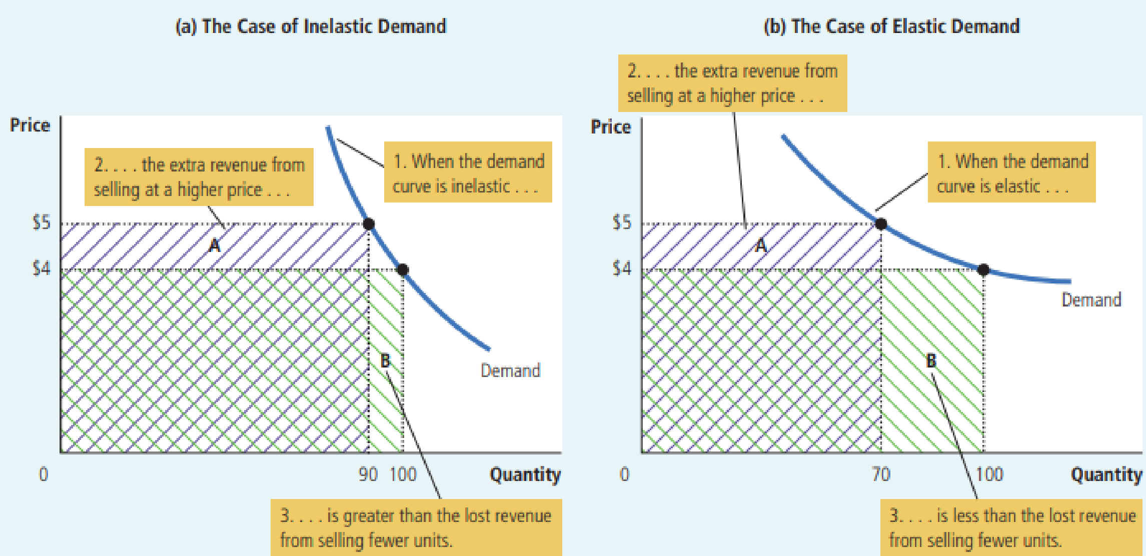
Book numbers — Inelastic: P rises \$4→\$5, Q falls 100→90. TR: \$400 → \$450 ↑ (Area A > Area B)

Book numbers — Elastic: P rises \$4→\$5, Q falls 100→70. TR: \$400 → \$350 ↓ (Area A < Area B)

FIGURE 3

How Total Revenue Changes When Price Changes

The impact of a price change on total revenue (the product of price and quantity) depends on the elasticity of demand. In panel (a), the demand curve is inelastic. In this case, an increase in the price leads to a decrease in quantity demanded that is proportionately smaller, so total revenue increases. Here an increase in the price from \$4 to \$5 causes the quantity demanded to fall from 100 to 90. Total revenue rises from \$400 to \$450. In panel (b), the demand curve is elastic. In this case, an increase in the price leads to a decrease in quantity demanded that is proportionately larger, so total revenue decreases. Here an increase in the price from \$4 to \$5 causes the quantity demanded to fall from 100 to 70. Total revenue falls from \$400 to \$350.



5-1f Elasticity Along a Linear Demand Curve

THE BIG SURPRISE

A straight (linear) demand curve has a CONSTANT SLOPE — but its ELASTICITY CHANGES at every point!
Most students assume: 'straight line = constant elasticity.' This is WRONG.
WHY? Slope = ratio of absolute changes (ΔP/ΔQ). Elasticity = ratio of PERCENTAGE changes (%ΔQd / %ΔP). These are different things!

WHY ELASTICITY CHANGES ALONG A STRAIGHT LINE

AT THE TOP (high P, low Q): A $1 price rise is a SMALL % of a high price. A 2-unit qty drop is a LARGE % of a small qty. So %ΔQd > %ΔP → ELASTIC.
AT THE BOTTOM (low P, high Q): A $1 price rise is a LARGE % of a low price. A 2-unit qty drop is a SMALL % of a large qty. So %ΔQd < %ΔP → INELASTIC.

THE THREE-ZONE RULE FOR LINEAR DEMAND CURVES

UPPER HALF of the curve → ELASTIC (E > 1)
EXACT MIDPOINT of the curve → UNIT ELASTIC (E = 1)
LOWER HALF of the curve → INELASTIC (E < 1)

TOTAL REVENUE IS MAXIMIZED at the midpoint (unit elastic). TR rises as price falls along the elastic upper half, then falls further along the inelastic lower half.

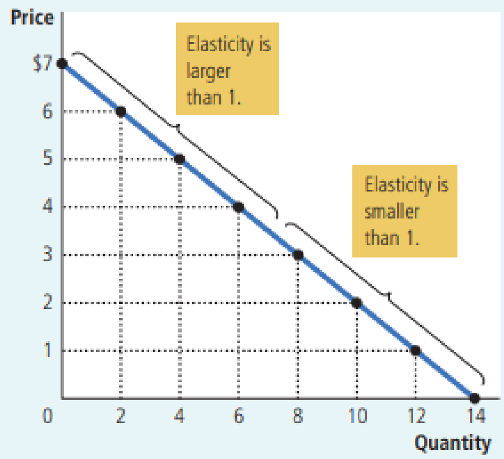


FIGURE 4

Elasticity along a Linear Demand Curve
The slope of a linear demand curve is constant, but its elasticity is not. The price elasticity of demand is calculated using the demand schedule in the table and the midpoint method. At points with a low price and high quantity, the demand curve is inelastic. At points with a high price and low quantity, the demand curve is elastic.

Price	Quantity	Total Revenue (Price × Quantity)	Percentage Change in Price	Percentage Change in Quantity	Elasticity	Description
$7	0	$0	15	200	13.0	Elastic
6	2	12	18	67	3.7	Elastic
5	4	20	22	40	1.8	Elastic
4	6	24	29	29	1.0	Unit elastic
3	8	24	40	22	0.6	Inelastic
2	10	20	67	18	0.3	Inelastic
1	12	12	200	15	0.1	Inelastic
0	14	0				

Data Table: Elasticity & TR Along a Linear Demand Curve

Price	Quantity	TR = P×Q	Elasticity	Zone
$7	0	$0	13.0	Elastic
$6	2	$12	3.7	Elastic
$5	4	$20	1.8	Elastic
$4	**6**	**$24**	**1.0**	**Unit Elastic ← TR MAX**
$3	8	$24	0.6	Inelastic
$2	10	$20	0.3	Inelastic
$1	12	$12	0.1	Inelastic

Note: Data from Figure 4 (Mankiw). TR is maximized at $24 where elasticity = 1.

5-1g Other Demand Elasticities

Income Elasticity of Demand (YED)

📐 **INCOME ELASTICITY FORMULA**

`YED = % Change in Quantity Demanded ÷ % Change in Income`

`Positive (+)` NORMAL good — buy MORE when income rises
`Negative (−)` INFERIOR good — buy LESS when income rises
`Large positive` LUXURY good (restaurants, holidays, diamonds)
`Small positive` NECESSITY (food, clothing, basic goods)

Cross-Price Elasticity of Demand (XED)

📐 **CROSS-PRICE ELASTICITY FORMULA**

`XED = %ΔQd of Good 1 ÷ %ΔP of Good 2`

`Positive (+)` SUBSTITUTES — goods that compete (Pepsi & Coke, butter & margarine)
`Negative (−)` COMPLEMENTS — goods used together (computers & software, cars & petrol)
`Zero (0)` UNRELATED goods — no connection

📈 Income Elasticity (YED)

- Formula: %ΔQd ÷ %ΔIncome
- Positive → Normal good (buy more when richer)
- Negative → Inferior good (buy less when richer)
- Small positive → Necessity (food, clothing)
- Large positive → Luxury (caviar, diamonds)

🔄 Cross-Price Elasticity (XED)

- Formula: %ΔQd Good1 ÷ %ΔP Good2
- Positive → Substitutes (Pepsi & Coke)
- Negative → Complements (cars & petrol)
- Zero → Unrelated goods
- Measures how two goods are RELATED

Elasticity	Sign	Meaning	Example
YED	Positive (+)	Normal good	Restaurants, cars, holidays
YED	Negative (−)	Inferior good	Bus rides, instant noodles
XED	Positive (+)	Substitutes	Pepsi & Coke
XED	Negative (−)	Complements	Computers & software

⚠ **COMMON CONFUSION — INFERIOR ≠ BAD QUALITY**

An INFERIOR GOOD is NOT a low-quality product.
It means: when your income goes UP, you buy LESS of it (because you switch to better alternatives).

Example: Bus rides. When people get richer, they buy cars. Bus rides = inferior good — not because buses are bad, but because richer people switch away from them.

💡 **REAL-LIFE EXAMPLES**

YED: Income doubles → you eat at restaurants 3× more (luxury, large YED) but buy only slightly more rice (necessity, small YED).
XED: Price of Coke rises → people buy more Pepsi (positive XED = substitutes). Price of printers rises → people buy less ink (negative XED = complements).

5-2 Price Elasticity of Supply (PES)

📖 **EASY DEFINITION**

Price Elasticity of Supply (PES) measures how much the QUANTITY SUPPLIED changes when the PRICE changes.
When price rises, do sellers produce a LOT more or just a LITTLE more?

📐 **PRICE ELASTICITY OF SUPPLY FORMULA**

PES = % Change in Quantity Supplied ÷ % Change in Price

PES > 1	Elastic supply — sellers respond a lot
PES < 1	Inelastic supply — sellers barely change quantity
PES = 0	Perfectly inelastic — fixed supply (beachfront land)
PES = ∞	Perfectly elastic — quantity adjusts completely

5-2a Key Determinant: Production Flexibility

🏖️ **Inelastic Supply (PES < 1)**

- Cannot produce more easily
- Fixed inputs (beachfront land, rare art)
- Short run — factory capacity is fixed
- Supply curve is steep
- Example: Beachfront land, Picasso paintings

🏭 **Elastic Supply (PES > 1)**

- Can expand production easily
- Manufactured goods (books, TVs, cars)
- Long run — build new factories, hire more
- Supply curve is flat/gentle
- Example: Books, electronics, clothing

Short Run vs Long Run for Supply

Time Period	PES	Why?
Short Run	Low — Inelastic	Factories have fixed capacity. Cannot quickly expand or shrink. Supply barely responds to price changes.
Long Run	High — Elastic	New factories built, new firms enter, old ones exit. Quantity supplied can change substantially over time.

5-2b Computing PES — Worked Example: Milk Farmers

STEP-BY-STEP CALCULATION

Milk price rises from $2.85 to $3.15. Production rises from 9,000 to 11,000 gallons.

Step 1 — Midpoint price: (2.85+3.15) ÷ 2 = $3.00
Step 2 — %ΔP: (3.15−2.85) ÷ 3.00 × 100 = 10%
Step 3 — Midpoint qty: (9,000+11,000) ÷ 2 = 10,000
Step 4 — %ΔQs: (11,000−9,000) ÷ 10,000 × 100 = 20%
Step 5 — PES = 20% ÷ 10% = 2 → Elastic supply (qty responds TWICE as much as price)

5-2c The 5 Types of Supply Curves

Type	PES	Curve Shape	Example
Perfectly Inelastic	E = 0	Vertical ↕	Beachfront land, original artworks
Inelastic	E < 1	Steep upward slope	Farm crops in the short run
Unit Elastic	E = 1	Moderate slope	Some manufactured goods
Elastic	E > 1	Gentle/flat slope	Books, cars, electronics
Perfectly Elastic	E = ∞	Horizontal ↔	Firms in perfectly competitive market

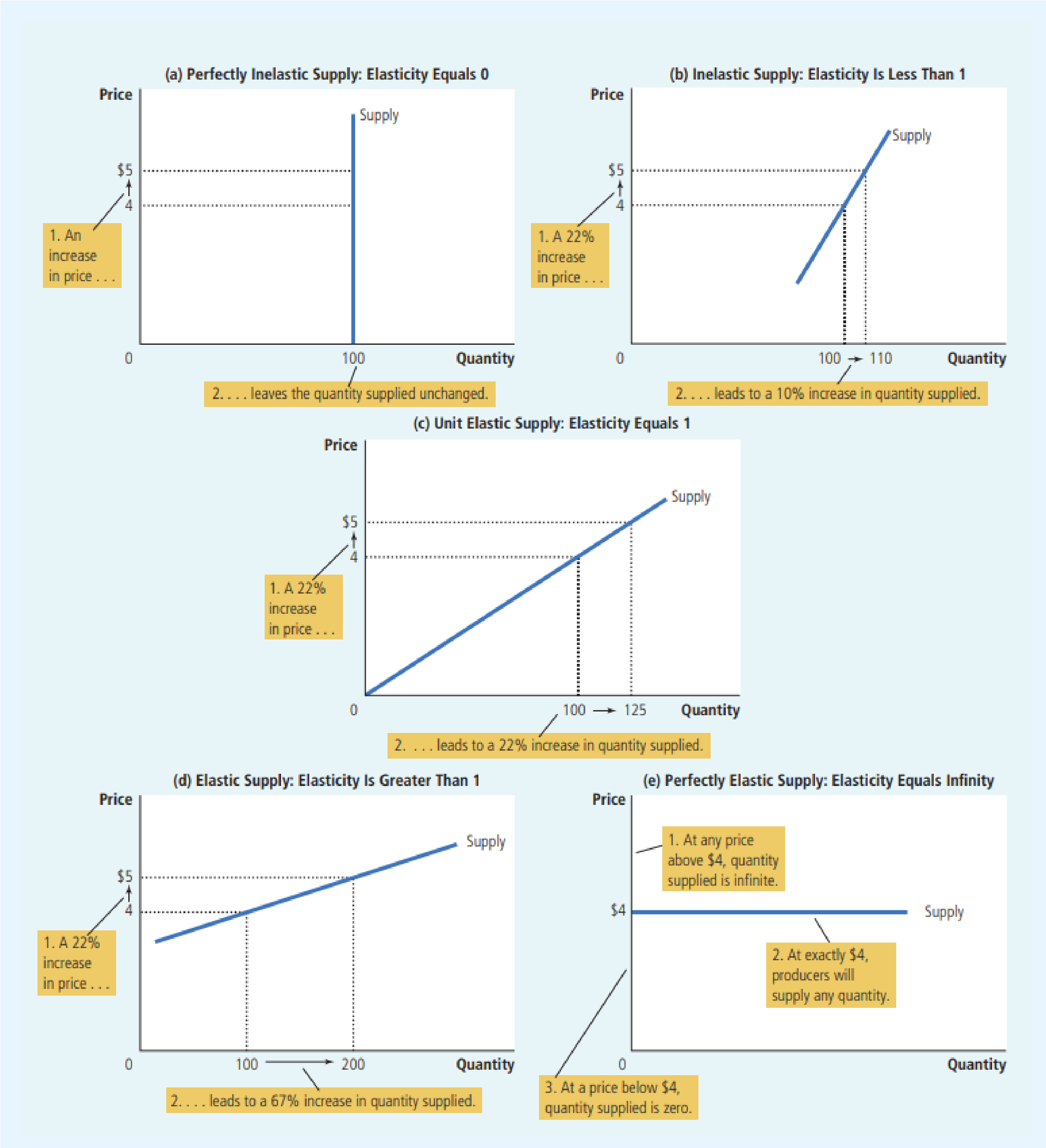
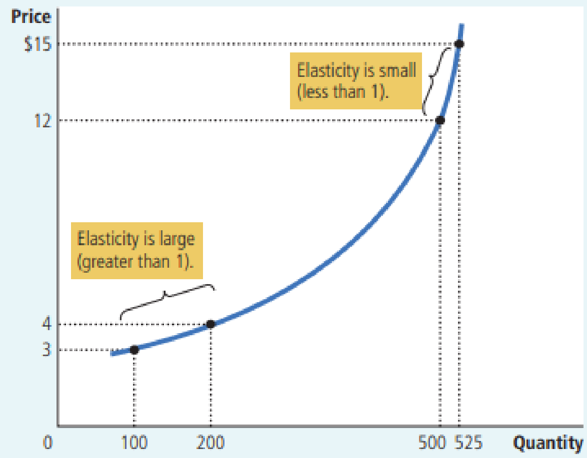

(a) Perfectly Inelastic Supply: Elasticity Equals 0
Price
Supply
$5
4
1. An increase in price . . .
0
100
Quantity
2. . . . leaves the quantity supplied unchanged.
(b) Inelastic Supply: Elasticity Is Less Than 1
Price
Supply
$5
4
1. A 22% increase in price . . .
0
100
110
Quantity
2. . . . leads to a 10% increase in quantity supplied.
(c) Unit Elastic Supply: Elasticity Equals 1
Price
Supply
$5
4
1. A 22% increase in price . . .
0
100
125
Quantity
2. . . . leads to a 22% increase in quantity supplied.
(d) Elastic Supply: Elasticity Is Greater Than 1
Price
Supply
$5
4
1. A 22% increase in price . . .
0
100
200
Quantity
2. . . . leads to a 67% increase in quantity supplied.
(e) Perfectly Elastic Supply: Elasticity Equals Infinity
Price
1. At any price above $4, quantity supplied is infinite.
$4
Supply
2. At exactly $4, producers will supply any quantity.
0
Quantity
3. At a price below $4, quantity supplied is zero.

FIGURE 6

How the Price Elasticity of Supply Can Vary

Because firms often have a maximum capacity for production, the elasticity of supply may be very high at low levels of quantity supplied and very low at high levels of quantity supplied. Here an increase in price from $3 to $4 increases the quantity supplied from 100 to 200. Because the 67 percent increase in quantity supplied (computed using the midpoint method) is larger than the 29 percent increase in price, the supply curve is elastic in this range. By contrast, when the price rises from $12 to $15, the quantity supplied rises only from 500 to 525. Because the 5 percent increase in quantity supplied is smaller than the 22 percent increase in price, the supply curve is inelastic in this range.



⚠ **PED VS PES — DON'T MIX THESE UP**

PED (Demand): affects BUYERS/CONSUMERS | Curve slopes DOWNWARD | Determined by substitutes, necessity, time

PES (Supply): affects SELLERS/PRODUCERS | Curve slopes UPWARD | Determined by production flexibility, time

BOTH are more elastic in the LONG RUN | BOTH always reported as positive numbers

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5-3a Application 1: Can Good News for Farming Be Bad News for Farmers?

📖 **THE SURPRISING QUESTION**

Scientists invent a new wheat hybrid that increases yield by 20%. Is this good or bad news for farmers?

SURPRISING ANSWER: It can be BAD news for farmers' total revenue!

Step-by-Step Analysis

1 **New hybrid → Supply increases**
Farmers produce 20% more wheat at any price → Supply curve shifts RIGHT ($S_1 \rightarrow S_2$)

2 **Demand stays the same**
Consumers' desire to buy wheat doesn't change because of a new farming method → Demand curve unchanged

3 **New equilibrium: Qty ↑, Price ↓**
Book example: Qty sold rises 100→110. Price falls $3→$2

4	**Demand for wheat is INELASTIC** Wheat/food has few substitutes and is a necessity → PED < 1

5	**Total Revenue FALLS!** Old TR = \$3 × 100 = \$300. New TR = \$2 × 110 = \$220. Revenue fell by \$80!

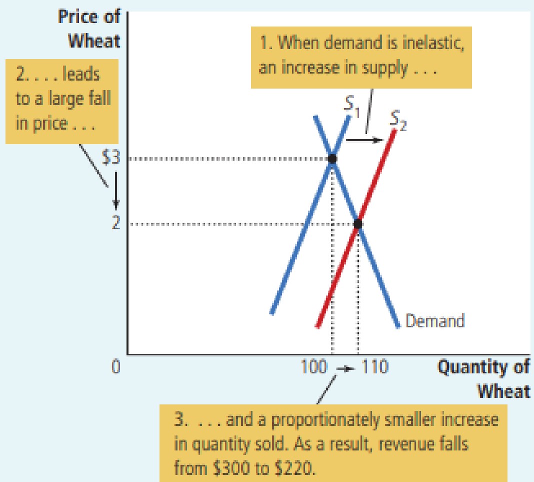


FIGURE 7

An Increase in Supply in the Market for Wheat

When an advance in farm technology increases the supply of wheat from S_1 to S_2, the price of wheat falls. Because the demand for wheat is inelastic, the increase in the quantity sold from 100 to 110 is proportionately smaller than the decrease in the price from \$3 to \$2. As a result, farmers' total revenue falls from \$300 (\$3 × 100) to \$220 (\$2 × 110).

WHY DON'T FARMERS REFUSE TO USE THE NEW TECHNOLOGY?

Each individual farmer is a tiny part of the whole wheat market. For any given market price, it is always rational for ONE farmer to grow more using the new hybrid — she earns more.

But when ALL farmers do this together, total supply surges, price crashes, and everyone earns less.

This is a classic collective action problem — individually rational behavior leads to a collectively bad outcome.

KEY EXAM POINTS

US farm employment dropped from 17% of the workforce in 1950 to just 2% today — technology boosted supply, inelastic food demand drove revenues down, millions left farming.

Government farm programs sometimes pay farmers NOT to plant crops — restricting supply keeps prices high. With inelastic demand: lower supply = higher total revenue for farmers.

What helps farmers (restricted supply, higher prices) HURTS consumers (higher food prices). Always a trade-off.

EXAM TRAP

Students assume 'more production = more revenue.' With INELASTIC demand, the opposite is true.

Always ask: What is the elasticity of demand? That determines whether a supply increase helps or hurts total revenue.

5-3b Application 2: Why Did OPEC Fail to Keep Oil Prices High?

BACKGROUND

OPEC (Organization of Petroleum Exporting Countries) agreed to reduce oil supply to raise prices and increase revenues. It worked short-run — but failed long-run. Why? Elasticity changes over time!
Key timeline: 1973-74: oil price up 50%+. 1979-81: price doubled again. 1982-85: price fell 10%/year. 1986: cartel collapsed, price plunged 45%. By 1990: back to 1970 level.

Short Run — OPEC Succeeds

- Supply inelastic (can't quickly find new oil)
- Demand inelastic (can't quickly change cars/habits)
- Both curves are STEEP
- OPEC cuts supply → supply shifts LEFT
- Steep curves → LARGE price increase
- OPEC earns much more even selling less oil

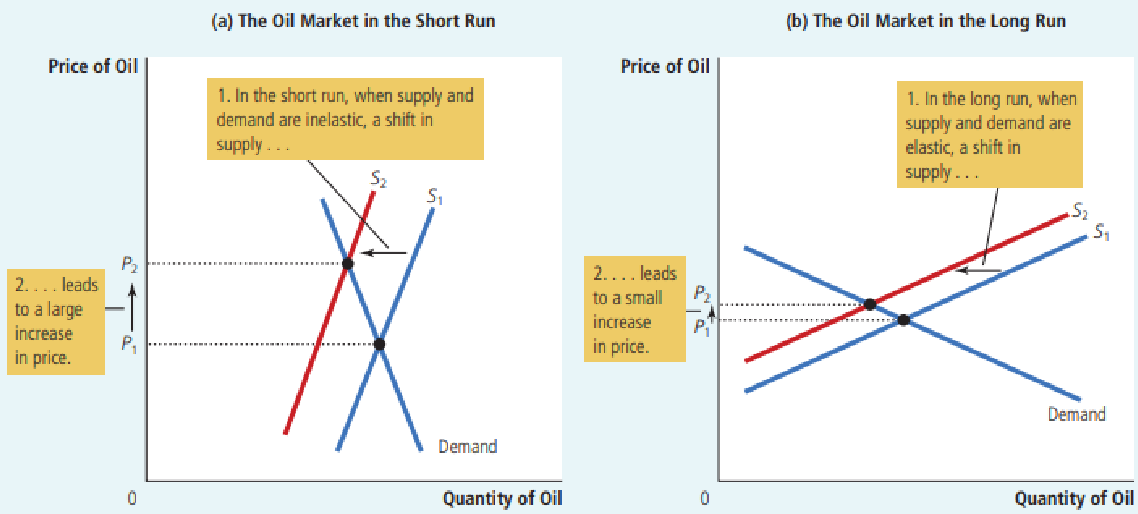
Long Run — OPEC Fails

- Supply elastic (fracking, new oil fields explored)
- Demand elastic (fuel-efficient cars, conservation)
- Both curves become FLAT
- Same supply cut → SMALLER price rise
- Revenue doesn't increase much
- OPEC members defect → cartel breaks down

When the supply of oil falls, the response depends on the time horizon. In the short run, supply and demand are relatively inelastic, as in panel (a). Thus, when the supply curve shifts from S_1 to S_2, the price rises substantially. In the long run, however, supply and demand are relatively elastic, as in panel (b). In this case, the same size shift in the supply curve (S_1 to S_2) causes a smaller increase in the price.

FIGURE 8

A Reduction in Supply in the World Market for Oil



KEY LESSON (DIRECT QUOTE FROM BOOK)

"Raising prices is easier in the short run than in the long run."
In markets where short-run elasticity is low but long-run elasticity is high, any price-raising strategy gradually loses power as producers and consumers adapt.
This applies to any cartel, monopoly, or government price-fixing scheme.

5-3c Application 3: Does Drug Interdiction Increase or Decrease Crime?

🕮 **THE POLICY QUESTION**

The government spends billions on drug interdiction (stopping drugs from entering the country, arresting suppliers). Does this reduce crime — or actually INCREASE it?
The answer depends entirely on the ELASTICITY OF DEMAND for drugs.

🚔 Policy A: Drug Interdiction (Attack Supply)

- Arrests suppliers → Supply shifts LEFT (S_1→S_2)
- Price of drugs RISES
- Quantity of drugs FALLS slightly
- Drug demand is INELASTIC (addicts can't easily stop)
- Total drug spending RISES (inelastic = P×Q up)
- Addicts need MORE money → steal MORE
- → CRIME INCREASES!

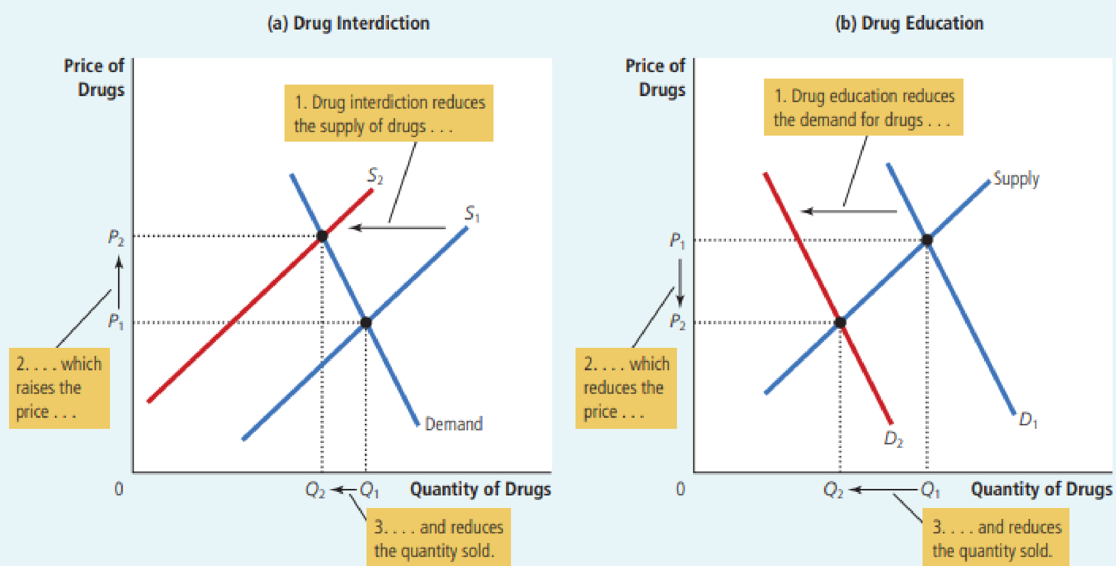
📚 Policy B: Drug Education (Attack Demand)

- Reduces desire → Demand shifts LEFT (D_1→D_2)
- Both price AND quantity FALL
- Total drug spending FALLS
- Addicts need LESS money
- → Less theft, less robbery
- → CRIME DECREASES! ✓
- Also reduces drug use simultaneously

FIGURE 9

Policies to Reduce the Use of Illegal Drugs

Drug interdiction reduces the supply of drugs from S_1 to S_2, as in panel (a). If the demand for drugs is inelastic, then the total amount paid by drug users rises, even as the amount of drug use falls. By contrast, drug education reduces the demand for drugs from D_1 to D_2, as in panel (b). Because both price and quantity fall, the amount paid by drug users falls.



Factor	Interdiction (Supply ↓)	Education (Demand ↓)
Which curve shifts?	Supply — leftward	Demand — leftward
Effect on Price	RISES	FALLS
Effect on Quantity	FALLS slightly	FALLS
Total Drug Spending	RISES (inelastic demand)	FALLS
Effect on Crime	May INCREASE	DECREASES

SHORT RUN VS LONG RUN CAVEAT

SHORT RUN: Drug demand is inelastic (existing addicts won't stop due to higher prices). Interdiction → higher crime. Education → lower crime.
LONG RUN: Higher prices may deter young people from experimenting. Over time demand becomes more elastic → fewer new addicts → interdiction may eventually reduce crime.
Both sides of the policy debate have valid economic arguments. The book presents both views honestly.

Full Chapter Summary

Elasticity — Big Idea

Measures responsiveness. Tells us HOW MUCH quantity changes when price, income, or another good's price changes. Goes beyond just 'up or down.'

PED Formula

PED = %ΔQd ÷ %ΔP. Use midpoint method for consistency. Elastic >1 | Inelastic <1 | Unit =1. Always positive by convention.

4 Determinants of PED

Substitutes (more = elastic) · Necessity/Luxury (luxury = elastic) · Market definition (narrow = elastic) · Time (long run = elastic). Memory: SNMT.

Midpoint Method

Use average of two values as base. Avoids direction problem. Gives same answer whether going A→B or B→A.

5 Types of Demand Curves

Perfectly inelastic (vertical, E=0) → Inelastic → Unit elastic → Elastic → Perfectly elastic (horizontal, E=∞). Flatter = more elastic.

Total Revenue Rule

Inelastic: P & TR same direction. Elastic: P & TR opposite. Unit elastic: TR unchanged. TR maximized at unit elastic midpoint of linear curve.

Income Elasticity (YED)

Positive = normal good. Negative = inferior good. Large positive = luxury. Small positive = necessity. Formula: %ΔQd ÷ %ΔIncome.

Cross-Price Elasticity (XED)

Positive = substitutes (Pepsi & Coke). Negative = complements (computers & software). Formula: $\%\Delta Qd_1 \div \%\Delta P_2$.

Price Elasticity of Supply

PES = %ΔQs ÷ %ΔP. Depends on production flexibility. Always more elastic in the long run. Vertical = perfectly inelastic (E=0).

Application 1: Farming

Better tech → supply↑ → price↓. With inelastic food demand → TR falls! Good tech news = bad revenue news for farmers as a group.

Application 2: OPEC Oil

Supply restriction works short run (inelastic). Fails long run as market becomes elastic. Key: raising prices easier short-run than long-run.

Application 3: Drug Policy

Interdiction (supply↓) → higher price → more crime (inelastic demand). Education (demand↓) → lower price AND qty → less crime. Education wins!

Ultra-Short Revision Sheet

Read this 5 minutes before your exam. Only the most critical facts.

⚡ **CRITICAL FACTS — READ BEFORE EXAM**

- Elasticity = measure of how much quantity responds to changes in price/income/other prices
- PED = %ΔQd ÷ %ΔP | Elastic > 1 | Inelastic < 1 | Unit = 1 | Always positive
- 4 Determinants of PED: Substitutes · Necessity/Luxury · Market Definition · Time Horizon
- Midpoint method = use average of two values as base → same answer both directions
- TR rule — Inelastic: Price & TR move SAME way | Elastic: Price & TR move OPPOSITE ways
- Linear demand: Upper half = elastic | Midpoint = unit elastic | Lower half = inelastic
- TR is MAXIMIZED at the unit elastic midpoint of a linear demand curve
- YED positive = normal good | YED negative = inferior good (not bad quality!)
- XED positive = substitutes | XED negative = complements
- PES = %ΔQs ÷ %ΔP | More elastic in long run | Flexible production = elastic supply
- Farming: Better tech → supply↑ → price↓ → inelastic demand → TR↓ (bad for farmers!)
- OPEC: Short run inelastic → big price rise. Long run elastic → small price rise. Cartel power fades.
- Drug interdiction: Supply↓ → price↑ → inelastic drug demand → total spending↑ → MORE crime!
- Drug education: Demand↓ → both price and qty↓ → total spending↓ → LESS crime ✓
- Memory trick (from the book!): 'Inelastic curves look like the letter I' (vertical = I shape)

All Formulas at a Glance

Formula	What It Calculates
PED = %ΔQd / %ΔP	Demand responsiveness to own price
PES = %ΔQs / %ΔP	Supply responsiveness to own price
YED = %ΔQd / %ΔIncome	Demand responsiveness to income
XED = $\%\Delta Qd_1 / \%\Delta P_2$	How Good 1 reacts to Good 2's price
TR = P × Q	Total Revenue (the rectangle on the graph)
Midpoint P = $(P_1+P_2)/2$	Base for midpoint % change calculation

Most Important Exam Questions & Model Answers

Short Questions (2–5 marks)

SHORT QUESTION

Q1: Define price elasticity of demand. What does it measure?

- PED measures how much the quantity demanded responds to a change in price.
- Formula: PED = %ΔQd ÷ %ΔP. Always report as a positive number.
- PED > 1 = elastic (responsive). PED < 1 = inelastic (not responsive). PED = 1 = unit elastic.

SHORT QUESTION

Q2: List and explain the four determinants of PED.

- (1) Availability of substitutes: more substitutes → more elastic (easy to switch).
- (2) Necessity vs. luxury: necessities (medicine) = inelastic; luxuries (diamonds) = elastic.
- (3) Market definition: narrowly defined markets are more elastic; broadly defined are more inelastic.
- (4) Time horizon: demand is more elastic in the long run (more time to adjust behavior).

SHORT QUESTION

Q3: What is the midpoint method and why is it preferred?

- The midpoint method calculates % changes using the average of initial and final values as the base.
- It is preferred because the standard method gives different answers going A→B vs B→A.
- Midpoint method gives the same answer in both directions — more consistent and accurate.

Broad / Essay Questions (8–15 marks)

BROAD QUESTION

Q4: Explain the relationship between PED and total revenue. Use examples and diagrams.

- Total revenue TR = P × Q. On a demand diagram, TR = the shaded P×Q rectangle.
- INELASTIC demand (PED < 1): Price rises → qty falls proportionately less → TR rises. Same direction. Example: P \$4→\$5, Q 100→90, TR \$400→\$450 ↑
- ELASTIC demand (PED > 1): Price rises → qty falls proportionately more → TR falls. Opposite direction. Example: P \$4→\$5, Q 100→70, TR \$400→\$350 ↓
- UNIT ELASTIC (PED = 1): TR unchanged when price changes.
- Draw Figure 3: show Area A (gained) vs Area B (lost). Inelastic: A > B. Elastic: A < B.
- Along a linear demand curve: TR maximized at the unit elastic midpoint.

BROAD QUESTION

Q5: 'Good news for farming can be bad news for farmers.' Explain with elasticity.

- New farming technology → supply curve shifts right ($S_1 \rightarrow S_2$). Demand unchanged.
- New equilibrium: price falls (\$3→\$2), quantity rises (100→110).
- Demand for food/wheat is INELASTIC (necessity, few substitutes).
- With inelastic demand: % fall in price > % rise in quantity.

- Therefore TR falls: Old TR = $300 → New TR = $220. Farmers earn LESS.
- Historical evidence: U.S. farm employment fell from 17% (1950) to 2% today.
- Policy: Governments pay farmers NOT to plant crops → restricts supply → raises TR (inelastic demand).

Graph Questions

GRAPH QUESTION

Q6: Draw and label the five types of demand curves. Explain each briefly.

- (a) Perfectly Inelastic (E=0): Draw a VERTICAL line. Label 'Demand.' At any price, qty stays at 100.
- (b) Inelastic (E<1): Steep downward line. Show price rise $4→$5 causes small qty fall (100→90).
- (c) Unit Elastic (E=1): Moderate slope. 22% price rise causes 22% qty fall.
- (d) Elastic (E>1): Gentle/flat slope. 22% price rise causes 67% qty fall.
- (e) Perfectly Elastic (E=∞): HORIZONTAL line at P=$4. Above $4 → qty = 0.
- Memory trick: Inelastic curve looks like the letter I (vertical). ✓

Analytical Questions

ANALYTICAL QUESTION

Q7: Explain why drug interdiction may increase crime. Compare with drug education.

- Drug interdiction: arrests suppliers → supply shifts left → price rises, qty falls slightly.
- Drug demand is inelastic (addicts cannot easily stop using when price rises).
- With inelastic demand: % price rise > % qty fall → total spending by users RISES.
- Addicts need more money → steal more → CRIME INCREASES.
- Drug education: reduces demand → demand shifts left → both price AND qty fall.
- Total drug spending FALLS → less need to steal → CRIME DECREASES. Better short-run policy!
- Long-run caveat: higher prices from interdiction may deter new users → less crime eventually.
- Draw Figure 9: panel (a) interdiction (supply shift, inelastic demand), panel (b) education (demand shift).

ANALYTICAL QUESTION

Q8: Why did OPEC succeed short-run but fail long-run? Explain with diagrams.

- OPEC cut production → supply shifted left in both short and long run (S_1→S_2).
- SHORT RUN: Both supply and demand are inelastic (steep curves). Same supply cut → LARGE price rise → OPEC revenues soar.
- LONG RUN: Both become more elastic (flat curves). Consumers buy fuel-efficient cars; new oil fields explored. Same supply cut → SMALL price rise → revenues barely improve.
- Draw Figure 8: two panels showing steep vs flat curves, same supply shift causing different price effects.
- Key lesson: 'Raising prices is easier in the short run than in the long run.' (direct book quote)

Common MCQ Concepts & Traps

These are the most-tested MCQ facts and the traps that catch students. Memorize all of them.

MCQ Fact 1: A life-saving medicine with NO close substitutes → SMALL (inelastic) PED.

⚠ Trap: 'No substitute' sounds extreme → students think elastic. WRONG! No substitute = can't switch = inelastic. (Book Q1, Answer: a)

MCQ Fact 2: Midpoint Method: Price \$8→\$12, Qty 110→90. Midpoint P=10, Q=100. %ΔP=40%, %ΔQ=20%. PED = 0.5 = 1/2.

⚠ Trap: Using old value (\$8) as base instead of midpoint → wrong answer. (Book Q2, Answer: b)

MCQ Fact 3: A linear downward-sloping demand curve → 'inelastic at some points and elastic at others.'

⚠ Trap: Students pick 'unit elastic' or 'constant elasticity.' A straight line does NOT have constant elasticity. (Book Q3, Answer: d)

MCQ Fact 4: Firms entering and exiting a market over time → the SUPPLY curve becomes more elastic.

⚠ Trap: Students pick 'demand curve more elastic.' Entry/exit affects SUPPLY, not demand. (Book Q4, Answer: c)

MCQ Fact 5: An increase in supply will DECREASE total revenue if demand is INELASTIC.

⚠ Trap: 'More supply = more revenue.' FALSE with inelastic demand — this is the farming application! (Book Q5, Answer: a)

MCQ Fact 6: Technology raises incomes and lowers prices. Spending rises if YED > 0 AND PED > 1.

⚠ Trap: Students pick 'one, one.' Income just needs positive YED. Price needs PED > 1 for lower price to RAISE spending. (Book Q6, Answer: b)

MCQ Fact 7: POSITIVE income elasticity = NORMAL good. NEGATIVE income elasticity = INFERIOR good.

⚠ Trap: 'Inferior' sounds like bad quality. It means buy LESS when income rises — nothing about quality.

MCQ Fact 8: POSITIVE cross-price elasticity = SUBSTITUTES. NEGATIVE cross-price elasticity = COMPLEMENTS.

⚠ Trap: Students swap these. Substitutes (+): price of A rises → buy more B. Complements (−): price of A rises → buy less B.

MCQ Fact 9: Price rises and total revenue ALSO rises → demand must be INELASTIC (PED < 1).

⚠ Trap: TR moved in same direction as price → always means inelastic demand. Opposite direction → elastic.

MCQ Fact 10: Drug interdiction reduces drug use BUT may increase drug-related CRIME.

⚠ Trap: Students assume 'less drug use = less crime.' Not true if inelastic demand causes total spending to rise.

MCQ Fact 11: **OPEC succeeded short-run because oil supply and demand are INELASTIC short-run.**

⚠ Trap: Students say OPEC failed because demand was elastic. NO — demand was INELASTIC initially. Elasticity INCREASED over time.

MCQ Fact 12: **Supply of beachfront land → Perfectly Inelastic (E=0). Cannot produce more at any price.**

⚠ Trap: Confusing perfectly inelastic supply with inelastic demand — they are different curves!

⚠ THE 5 BIGGEST EXAM TRAPS — NEVER FALL FOR THESE

Trap 1: 'Price rises → total revenue always rises' — FALSE when demand is ELASTIC!

Trap 2: 'A straight demand line has constant elasticity' — FALSE! Slope is constant, elasticity changes along the curve.

Trap 3: 'Drug interdiction always reduces crime' — NOT necessarily! With inelastic demand, it may actually INCREASE crime.

Trap 4: 'Inferior good = bad quality product' — FALSE! Inferior means you buy LESS of it when your income RISES.

Trap 5: 'Supply elasticity is the same short-run and long-run' — FALSE! Supply is always more elastic in the long run.